

SWARAJ LADKE

AI & Full Stack Developer

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CAREER SUMMARY:

AI & Full-Stack Engineer building production-grade web applications and conducting independent research into biologically-plausible, zero-forgetting neural architectures. Experienced across the full stack — from deployed VS Code extensions and SaaS platforms to novel deep learning systems achieving 92%+ continual learning retention. Oracle-certified in Generative AI and Cloud Engineering, with peer-reviewed research publications.

RELEVANT PROJECTS:

- **CODELIXER - VS CODE EXTENSION :** (Live Deployment).
Developed a TypeScript-based VS Code extension leveraging LLM APIs (Groq/OpenAI) to auto-correct code and provide real-time explanations. Implemented a context-aware correction engine, improving code accuracy by 40% in tests. (Technologies: TypeScript, Node.js, VS Code Extension API, OpenAI/Groq API)
- **HireFlow AI – Autonomous AI Recruitment & Proctoring Platform.**
Engineered an end-to-end, multi-service automated recruitment platform featuring an ATS resume screener, an AI-driven behavioral interviewer, and a locally-proctored live coding environment to evaluate software engineering candidates autonomously. (Tech Stack: React, Next.js, Python, Flask, SQLite, OpenCV, REST APIs, Micro-frontends)
- **AGNIS-DRIVE :** (Live Deployment).
Developed a full-stack cloud drive enabling secure file storage, organization, and sharing. Implemented real-time synchronization and cross-device access for seamless collaboration. Built for performance and scalability.
(Technologies: React 19, Next.js 15, Appwrite, TypeScript, Tailwind CSS, shadcn/ui)
- **AGNIS DESIGN STUDIO : (Live Deployment).**
Developed a full-stack web design platform with drag-and-drop editor and real-time collaboration, delivering fast, scalable, and intuitive workflows across devices. (Technologies: Next.js, TypeScript, Liveblocks, Fabric.js, Tailwind CSS, shadcn/ui)
- **AGNIS (AUTONOMOUS GATED NEURAL INFERENCE SYSTEM) | ONGOING RESEARCH**
Currently developing a biologically-plausible Deep Learning system focused on Zero-Forgetting Continual Learning. Architecting an Iterative Settlement Mechanism using Predictive Coding and Synchronized Aggressive Target Propagation (SNAP-ATP) to bypass traditional backpropagation, achieving initial milestones of 92% bilingual retention.
Tech Stack: Python, PyTorch, CUDA, Predictive Coding, Hebbian Learning.
Portfolio (All Projects & Live Demos): <https://swarajladke.github.io/portfolio>
GitHub (Source Codes): <https://github.com/swarajladke>

WORK EXPERIENCE :

TechnoHacks solutions Pvt.Ltd	Full Stack Web Developer Intern (Remote)	PUNE, INDIA
• Built and deployed responsive web applications using React.js, Flask, and REST APIs, improving performance and UX responsiveness by 25%. • Integrated third-party APIs and optimized front-end workflows using React hooks and modular components. • Collaborated with developers on debugging and CI/CD deployment pipelines through GitHub and Render.		Jun 2025 – July 2025

TECHNICAL SKILLS :

Languages: Python, JavaScript, TypeScript

Frontend: React, Next.js, Tailwind CSS, ShadCN, Fabric.js, Redux, Framer Motion, Vanta.js, Three.js, Xterm.js

AI/ML: PyTorch, TensorFlow, Hugging Face Transformers, CUDA, Predictive Coding, Hebbian Learning, Continual Learning, LangChain, NLP, LLMs, Scikit-learn, OpenCV

Backend & Databases: FastAPI, Flask, Node.js, MongoDB, PostgreSQL, MySQL, Supabase

Cloud & DevOps: AWS, Azure, Google Cloud, Vertex AI (GCP), Docker, Vercel, Render, GitHub, CI/CD

Tools & APIs: VS Code, Cursor, Windsurf, Antigravity, OpenAI API, Groq API, Postman, Vite, Monaco Editor, Appwrite, Firebase Auth, OAuth, Google Colab, Kaggle

EDUCATION :

- Masters of Computer Applications - 2024-2026
(Savitribai Phule Pune University)
- Bachelors of Computer Applications - 2021-2024
(Savitribai Phule Pune University)

CERTIFICATIONS AND PUBLICATIONS : (verification links on portfolio)

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Oracle APEX Cloud Certified Developer Professional
- AI & Career Empowerment - (Robert H. Smith Business School, University of Maryland, USA)
- AI for Work & Life - (University of North Florida, USA)
- Emoify (Research paper) : <https://zenodo.org/records/15752110>
- AI Augmented Digital Twin (Research paper) : <https://ijsci.com/index.php/home/article/view/744>